CentromereDB: A Pan-Eukaryotic Centromere Annotation Dataset for Training Genome Language Models

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1 What is a Centromere?

Centromeres are specialised chromosomal regions essential for accurate cell division, serving as the attachment sites where spindle fibres connect to chromosomes during mitosis and meiosis. Despite their fundamental importance for genome stability and inheritance, centromeres exhibit remarkable structural and sequence diversity across eukaryotic species[1]. In budding yeasts, centromeres are compact point centromeres spanning only $\sim$125 base pairs with defined sequence motifs[2]. In contrast, mammalian centromeres can sprawl megabases with highly repetitive satellite DNA sequences. Efforts to link the highly diverse centromere sequences, which can vary drastically even within a species, to the highly conserved function have so far failed. There are no universal sequence signatures that define centromeres across all species[1]. Consequently, due to their challenging annotation, centromeres are often excluded from whole-genome assemblies and downstream analysis. However, centromeres represent a large percentage of a genome, approximately 5–10% in the human genome[3], more than protein coding genes, making their omission a significant gap in our understanding of chromosome structure, function, and inheritance.

2 AI Task: The Training Data Problem for Centromeres

The lack of comprehensive centromere annotations across taxonomic diversity has created a critical gap in genomic training data that limits the development of truly whole-genome-aware AI systems. In current genome language models (gLMs), centromeres are systematically masked, excluded or weighted down in training datasets[4, 5, 6] due to their repetitive nature and poor annotation quality. This creates **systematic bias in genome language models** that prevents us from learning the true complexity and functional importance of large swaths of the genome.

CentromereDB will address this fundamental training data gap by providing the **first high-quality dataset of annotated centromeric sequences across eukaryotic diversity**, providing the scale needed for machine learning applications. Once gLMs are trained on complete genomes, including properly annotated centromeres, they will enable numerous downstream scientific applications.

3 Dataset Rationale

The fundamental bottleneck preventing centromere-aware genomic AI is that centromere annotations have never been applied systematically across taxonomic diversity (only in humans and a handful of model organisms). CentromereDB will provide the missing training data by delivering centromere annotations across the taxonomic breadth of eukaryotic life. Built upon the Darwin Tree of Life collection's 2,037 chromosome-level genome assemblies representing 1.77 trillion base pairs[7], the dataset will include precisely annotated centromeric boundaries, detailed classification

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of satellite DNA and transposon families within centromeres and metadata distinguishing different centromere architectures[8]. Importantly, this data has all been assembled in a highly standardised pipeline with extensive quality control, developed by the Darwin Tree of Life team, which would be absent from a meta-analysis or by combining data from multiple sources[7].

4 Acceleration Potential

Access to CentromereDB will enable the **training of the first generation of gLMs that understand complete chromosome architecture** (Appendix, **Fig. 1**). Complete genomic training data will unlock applications currently impossible with centromere-masked models, which will potentially transform our understanding of chromosome biology. Synthetic biology will benefit from AI systems that can design functional artificial chromosomes with predictable inheritance patterns[9]. Cancer research will gain tools for understanding how centromeric instabilities contribute to chromosomal aberrations and tumour evolution[10]. Agricultural biotechnology will be able to engineer crops with enhanced chromosomal stability and novel inheritance mechanisms[10, 11].

CentromereDB will establish standardised benchmarks for evaluating gLMs on complete genome understanding rather than artificially restricted sequence sets. This will drive improvements in model architectures and training procedures while providing researchers with **objective metrics for assessing whether their models truly understand genome organization** or are simply performing well on incomplete data.

5 Data Creation Pathway

Centromere annotation will employ a novel multi-modal pipeline integrating complementary detection approaches. K-mer-based methods (Panagram, `https://github.com/kjenike/panagram`) will identify short shared sequence across evolutionarily distant species. Repetitive sequence annotation tools (TRASH[12], EDTA[13], RepeatModeler[14]) will systematically classify and map centromeric repeats, including satellites and centromeric retrotransposons. The available chromatin conformation capture (Hi-C) will delineate three-dimensional chromatin organisation and identify the constrained, loop-forming regions characteristic of functional centromeres. Where available, epigenetic profiling through ChIP-seq for centromere-specific histone variants (CENP-A) will provide direct functional validation of computationally predicted centromeric boundaries.

6 Cost-Effective Strategies

This approach leverages existing high-quality genome assemblies from Darwin Tree of Life, eliminating the substantial costs typically associated with de novo sequencing and assembly generation. Computational requirements, while significant for multi-modal integration and validation, are supported through existing institutional HPC allocations and national grant computing allowances we have applied for ($\sim$500,000 computing hours and 50 TB), making the approach financially sustainable for large-scale comparative genomics.

7 Appendix

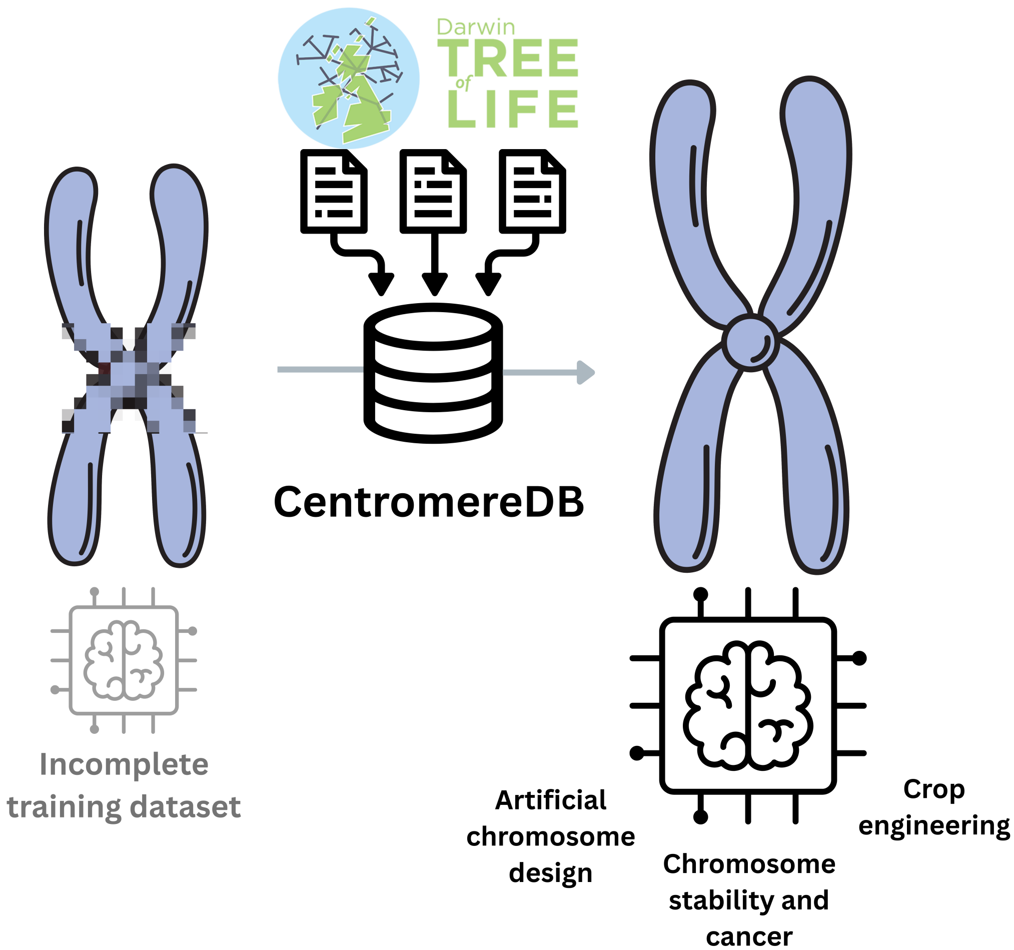

Figure 1: CentromereDB will be the first-of-its-kind database, providing accurate annotation of centromeres in a wide range of organisms. This would enable truly whole-genome-aware gLM training and subsequent applications.

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